Exploring Legacy FPGA Hardware for Telco Security: An Experimental GTP-U Monitoring Architecture on Xilinx Virtex-5

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ABSTRACT

The rapid evolution of mobile networks has increased the complexity of security monitoring across the radio, transport, and core network domains. In mobile-core environments, GTP-U traffic provides valuable information for observing tunnel activity, traffic behavior, and protocol characteristics, yet conventional software-based packet inspection places a significant processing burden on general-purpose host systems. This work explores an alternative approach through a hardware/software co-design experiment using a Digilent Genesys development board equipped with a Xilinx Virtex-5 LX50T FPGA.

The research introduces FALCON (FPGA-Accelerated Live Core Observation Node), an experimental FPGA-based observation architecture designed to process GTP-U traffic directly in programmable logic. The implemented hardware pipeline performs packet parsing, protocol identification, statistical accumulation, event generation, and telemetry packing before transferring compact observation data to a host-side collector. The host software subsequently decodes the hardware-generated telemetry and provides real-time monitoring through network interfaces and visualization components. A common telemetry format is used to maintain a deterministic interface between the RTL implementation and the software stack.

The main objective of the experiment is not to propose a production-ready mobile-core security appliance, but to investigate the practical feasibility of using a relatively constrained FPGA platform as a dedicated security observation engine. The evaluation therefore focuses on the behavior of the implemented RTL on physical Virtex-5 hardware, including packet-processing correctness, telemetry consistency, hardware/software interoperability, and the ability of the architecture to maintain continuous observation while transferring only compact security-relevant metadata to the host. The experimental results demonstrate that the proposed architecture can perform GTP-U-oriented observation on physical FPGA hardware and successfully deliver structured telemetry to a host monitoring system.

The findings suggest that programmable hardware can provide an additional observation layer for defense-in-depth in telco environments. Although the present prototype does not implement a complete collective-defense framework, the generated telemetry can serve as a foundation for subsequent security analytics, SIEM integration, anomaly detection, and privacy-aware exchange of security indicators between network domains. The study therefore demonstrates how hardware-level experimentation with legacy FPGA technology can contribute to the development of practical and layered security mechanisms for modern telecommunications networks.

1. Introduction

The security architecture of modern telecommunication networks increasingly depends on continuous visibility into traffic moving across distributed network elements. As mobile networks have evolved toward highly virtualized and programmable infrastructures, the volume and complexity of traffic that must be monitored have also increased. Security monitoring can no longer depend exclusively on perimeter inspection or host-based logging because important indicators of abnormal behavior may only become visible when traffic characteristics are observed directly inside the packet-processing path.

Within the mobile core, GTP-U is particularly relevant because it transports user-plane traffic through tunnels identified by Tunnel Endpoint Identifiers (TEIDs). Although the payload may remain opaque to many monitoring systems, the structure and behavior of GTP-U traffic provide useful information for security observation. Tunnel activity, packet rates, protocol characteristics, directionality, and unusual packet conditions can be converted into security telemetry without necessarily forwarding every packet to a general-purpose processor. This creates an opportunity to investigate whether part of the observation process can be performed directly in programmable hardware.

Field-Programmable Gate Arrays (FPGAs) provide an interesting platform for this purpose because packet-processing functions can be implemented as deterministic streaming logic. Instead of treating packet inspection as a sequence of software operations executed by a CPU, parsing and classification can be implemented as hardware pipelines that inspect incoming data as it traverses the device. Such an approach has been explored in various high-performance networking and hardware-security research domains, but its application to telco-oriented security telemetry remains an attractive area for experimentation, particularly when considering platforms that are no longer considered mainstream deployment hardware.

This research therefore takes a deliberately experimental approach. Rather than starting from a modern FPGA accelerator and evaluating only its maximum performance, the study investigates whether a comparatively constrained and much older FPGA platform can still be used to implement a functional security observation pipeline for mobile-core traffic. The experimental platform is a Digilent Genesys development board based on the Xilinx Virtex-5 LX50T FPGA. The original Genesys board provides a Virtex-5 LX50T device together with Gigabit Ethernet, DDR2 memory, programmable clocking, JTAG configuration, and expansion interfaces, making it suitable for hardware experimentation and network-oriented prototyping.

The choice of the Virtex-5 platform is intentional. Modern FPGA families provide considerably more resources and more advanced networking interfaces, but using an older device introduces an additional research question: how far can a constrained FPGA architecture be pushed when the objective is not full packet forwarding or line-rate switching, but security observation and telemetry extraction? The constraint changes the engineering problem. Logic resources, memory, clocking, interface availability, and software-tool compatibility become part of the experiment rather than hidden implementation details.

FALCON (FPGA-Accelerated Live Core Observation Node) was developed to investigate this question. The architecture places the first stage of GTP-U observation inside FPGA logic. Incoming packet data is processed by a hardware parser that identifies the relevant protocol fields and extracts information required by the monitoring pipeline. The parsed information is then passed to protocol classification and statistical logic, where counters and event-related information are accumulated before being converted into compact telemetry messages. Instead of sending complete packets to the host, the hardware produces structured metadata representing the observed state of the traffic.

The resulting architecture creates a clear hardware/software boundary. The FPGA is responsible for deterministic packet observation and first-stage aggregation, while the host software acts as a telemetry consumer and monitoring layer. This separation makes the experiment particularly useful from a security-research perspective because it allows the packet-processing function to be studied independently from visualization and higher-level analytics. It also provides a practical method for testing the same telemetry contract using simulation before connecting the software stack to physical FPGA hardware.

A central component of the FALCON implementation is the telemetry contract shared by the RTL and host software. The interface defines common message fields for message identification, versioning, and payload length, followed by structured telemetry types representing global statistics, per-TEID information, event information, and protocol-level statistics. The use of an explicit binary contract reduces ambiguity between the hardware and software domains and allows telemetry generated by the FPGA to be decoded deterministically by the host. More importantly, it enables hardware/software interoperability to be tested as part of the experiment rather than treated as an assumption.

The research consequently evaluates more than whether the FPGA design can be synthesized. A meaningful hardware security experiment must establish that the implemented logic behaves correctly after deployment to physical hardware, that generated telemetry remains consistent with

the expected protocol structure, and that the host can reliably consume the information produced by the device. For this reason, the FALCON experiment considers the complete path from traffic generation and packet observation to FPGA telemetry emission, network transport, host decoding, and visualization. The physical FPGA therefore becomes part of the measurement environment rather than merely a target for compilation.

This distinction is important because simulation alone cannot answer several practical questions associated with FPGA-based security instrumentation. Simulation can verify protocol parsing, state-machine behavior, and expected telemetry contents, but physical deployment introduces additional factors such as clock behavior, interface timing, actual configuration of I/O resources, implementation constraints, and interaction with external network equipment. Consequently, the use of the Genesys Virtex-5 board transforms FALCON from a purely conceptual packet-processing design into a hardware experiment in which the complete observation chain can be exercised.

From a telecommunications security perspective, the experiment also explores an alternative placement of security visibility. Conventional monitoring architectures commonly concentrate processing in software-based sensors, collectors, or centralized analytics platforms. FALCON instead introduces an observation layer directly inside programmable hardware. The purpose is not to replace software security systems, but to examine whether hardware can perform selected security-relevant transformations before information reaches the host. This can reduce the amount of raw traffic that needs to be transferred into subsequent processing stages and provides a potential architectural boundary between high-rate packet observation and higher-level security analysis.

This hardware-centric view is closely related to the principle of defense in depth. A robust security architecture should not depend on one inspection mechanism, one software process, or one centralized monitoring point. Hardware-level observation can provide an additional layer below host-based analytics, while the host can subsequently feed SIEM, anomaly-detection, or incident-response systems. In such an architecture, the FPGA does not become the complete security system; instead, it supplies an independent observation capability that can complement other controls.

The broader relevance becomes more apparent in the context of collective defense. The theme of IDSECCONF 2026, “Defense in Depth: The Rise of Collective Defense,” highlights the need for security mechanisms that operate across multiple layers and potentially across multiple domains. However, collective defense depends on the availability of trustworthy telemetry before intelligence can be correlated or shared. A hardware observation layer therefore represents an enabling component: it can generate structured security-relevant signals locally, while downstream systems

can decide which information should remain internal and which indicators can be shared with other security domains.

The contribution of this work is consequently centered on experimentation rather than the claim of a completed production platform. First, the study demonstrates the implementation of a GTP-U-oriented observation pipeline in RTL targeting a physical Xilinx Virtex-5 FPGA. Second, it establishes a hardware/software telemetry boundary that allows packet-processing logic and host-side monitoring components to be evaluated independently. Third, it examines the feasibility of using a constrained legacy FPGA platform as a dedicated observation layer for telco security. Finally, it discusses how the resulting telemetry can become a building block for layered security architectures in which local hardware observation contributes to larger monitoring and collective-defense mechanisms.

By focusing on the physical experimentation itself, this work treats the FPGA not merely as an implementation target but as a research instrument. The central question is therefore not whether an FPGA can theoretically process network traffic, but what security-relevant functionality can actually be implemented, observed, and validated on a real Virtex-5 platform under the constraints of an experimental telco environment. The results provide a starting point for further investigation into hardware-accelerated security telemetry, resource-aware telco monitoring, and the role of programmable hardware in defense-in-depth architectures.

2. Field-Programmable Gate Array (FPGA)

A Field-Programmable Gate Array (FPGA) is a type of programmable integrated circuit whose internal digital logic can be configured by the user after manufacturing. Unlike a conventional processor, which executes software instructions sequentially on a fixed processor architecture, an FPGA allows the designer to construct and modify the digital hardware itself according to the requirements of a particular application. This characteristic makes an FPGA fundamentally different from a CPU or a conventional Application-Specific Integrated Circuit (ASIC). In a CPU, the hardware architecture is predetermined and software controls how that hardware executes instructions, whereas in an FPGA, the hardware datapath, control logic, and interconnection structure can be configured to implement a specific digital function.

The internal structure of an FPGA generally consists of configurable logic blocks, programmable interconnection resources, memory elements, and dedicated hardware resources. Depending on the FPGA family, these resources may include Look-Up Tables (LUTs), flip-flops, Block RAM

(BRAM), Digital Signal Processing (DSP) blocks, clock-management resources, and high-speed I/O interfaces. LUTs are commonly used to implement combinational logic, while flip-flops provide sequential state storage. BRAM can be used for packet buffers, lookup tables, counters, or other stateful processing. These components can be connected through programmable routing resources to form a hardware datapath tailored to the application.

The programmability of an FPGA is typically described using a Hardware Description Language (HDL), such as VHDL or Verilog. Rather than writing a sequence of instructions for a processor, the designer describes the behavior and structure of the required digital circuit. The HDL design is then synthesized into a logical representation, mapped onto the available FPGA resources, and routed through the device's programmable interconnect. The resulting configuration is commonly stored in a bitstream that is loaded into the FPGA. After configuration, the device behaves according to the implemented hardware design. This process makes FPGA development closer to digital hardware engineering than conventional software development.

One of the main advantages of FPGA-based processing is its ability to perform operations in parallel. Multiple packet-processing functions can operate simultaneously because they are implemented as independent hardware stages. For example, a network-processing design can parse packet headers, classify protocols, update counters, and generate metadata through different pipeline stages without requiring the processor to execute each operation sequentially. This property is particularly relevant to high-speed networking, Software-Defined Radio (SDR), signal processing, industrial control, and security applications where deterministic processing and low-latency observation are important.

FPGAs also provide a useful middle ground between general-purpose processors and ASICs. CPUs provide excellent flexibility because their behavior can be changed by software, but packet-intensive or signal-processing workloads may consume significant computational resources when implemented entirely in software. ASICs can provide very high performance and efficiency, but their hardware functionality is effectively fixed after fabrication and changing the design requires a new manufacturing cycle. An FPGA retains a substantial degree of hardware flexibility while allowing specialized datapaths to be implemented without fabricating a new chip. This makes FPGA platforms particularly useful for research and prototyping, where the hardware architecture may need to be modified repeatedly during experimentation.

From a network-security perspective, these characteristics allow an FPGA to be used as an observation or inspection engine positioned close to the data path. Instead of forwarding every

packet to a host CPU, selected header fields and traffic characteristics can be extracted directly in hardware. A security-oriented FPGA design can, for example, identify protocol headers, track tunnel identifiers, maintain traffic counters, recognize predefined events, and generate compact telemetry for a host system. The FPGA therefore does not necessarily need to perform complete deep-packet inspection. It can instead perform deterministic first-stage processing and forward security-relevant metadata to higher-level software.

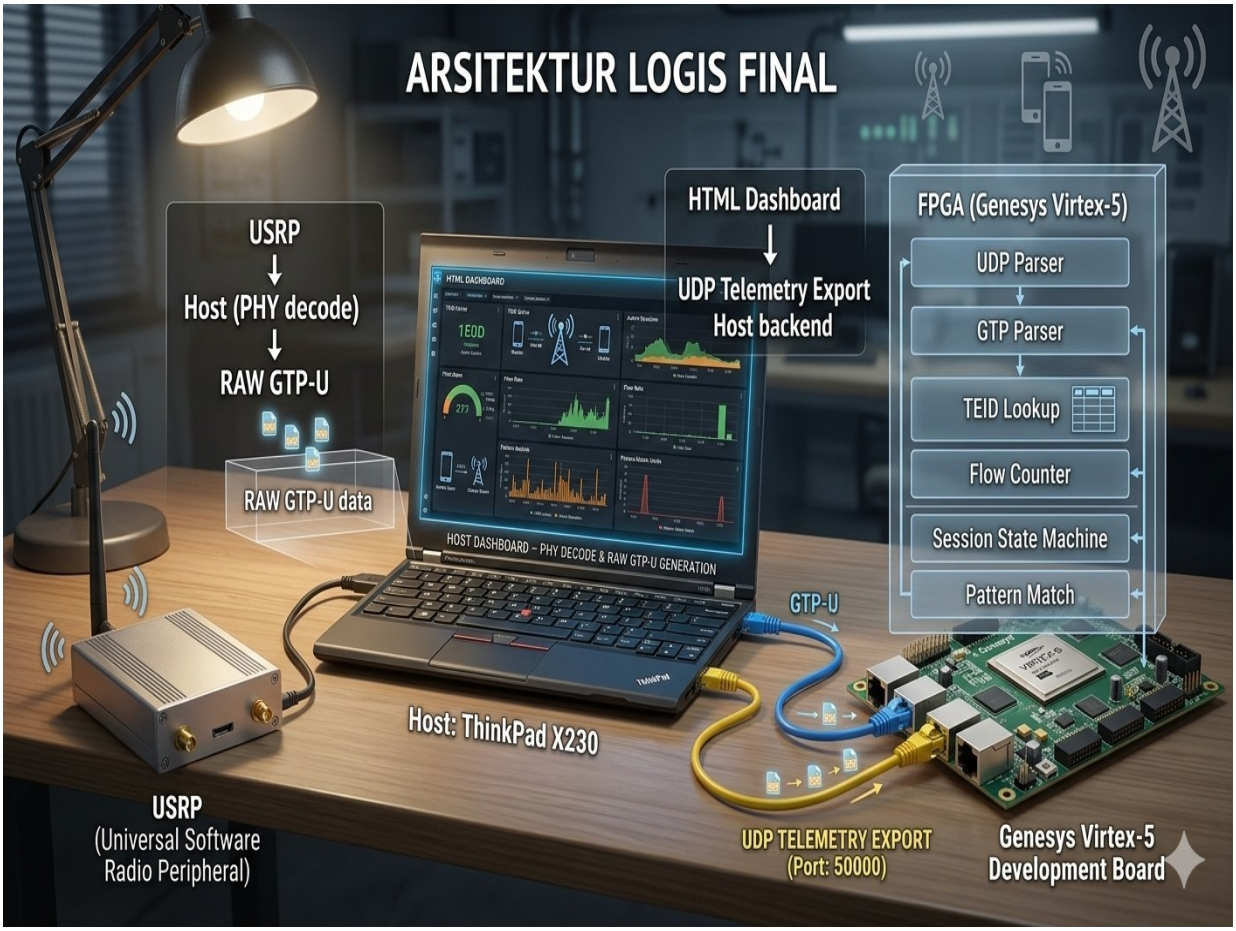
This architectural approach is particularly relevant to telecommunications networks. Mobile-core traffic can contain large numbers of packets and sessions, making continuous software-based observation potentially expensive. A programmable hardware layer can perform initial parsing and aggregation before the resulting information is sent to a host collector, SIEM, anomaly-detection engine, or other security analytics platform. In such a design, the FPGA becomes one layer in a broader defense-in-depth architecture rather than functioning as a complete security system by itself.

In this research, the FPGA is used as the primary experimental platform for investigating hardware-assisted observation of GTP-U traffic. The selected hardware is a Digilent Genesys development board based on the Xilinx Virtex-5 LX50T FPGA. The use of this platform is particularly interesting because it demonstrates that hardware-assisted telco security experimentation does not necessarily require a modern, high-end FPGA. Instead, a relatively constrained FPGA can be studied as a programmable environment in which packet parsing, protocol classification, statistical processing, and telemetry generation are implemented as hardware logic.

The role of the FPGA in the FALCON experiment can therefore be summarized as a programmable packet-observation engine. Incoming traffic is processed through hardware logic, relevant information is extracted and aggregated, and the resulting telemetry is transferred to the host for further processing and visualization. This separation between hardware-level observation and software-level analysis provides an experimental foundation for evaluating how programmable hardware can contribute to scalable and layered security monitoring in telecommunications networks.

3. METHODOLOGY AND SYSTEM ARCHITECTURE

3.1 System Overview



Ilustrasi architecture

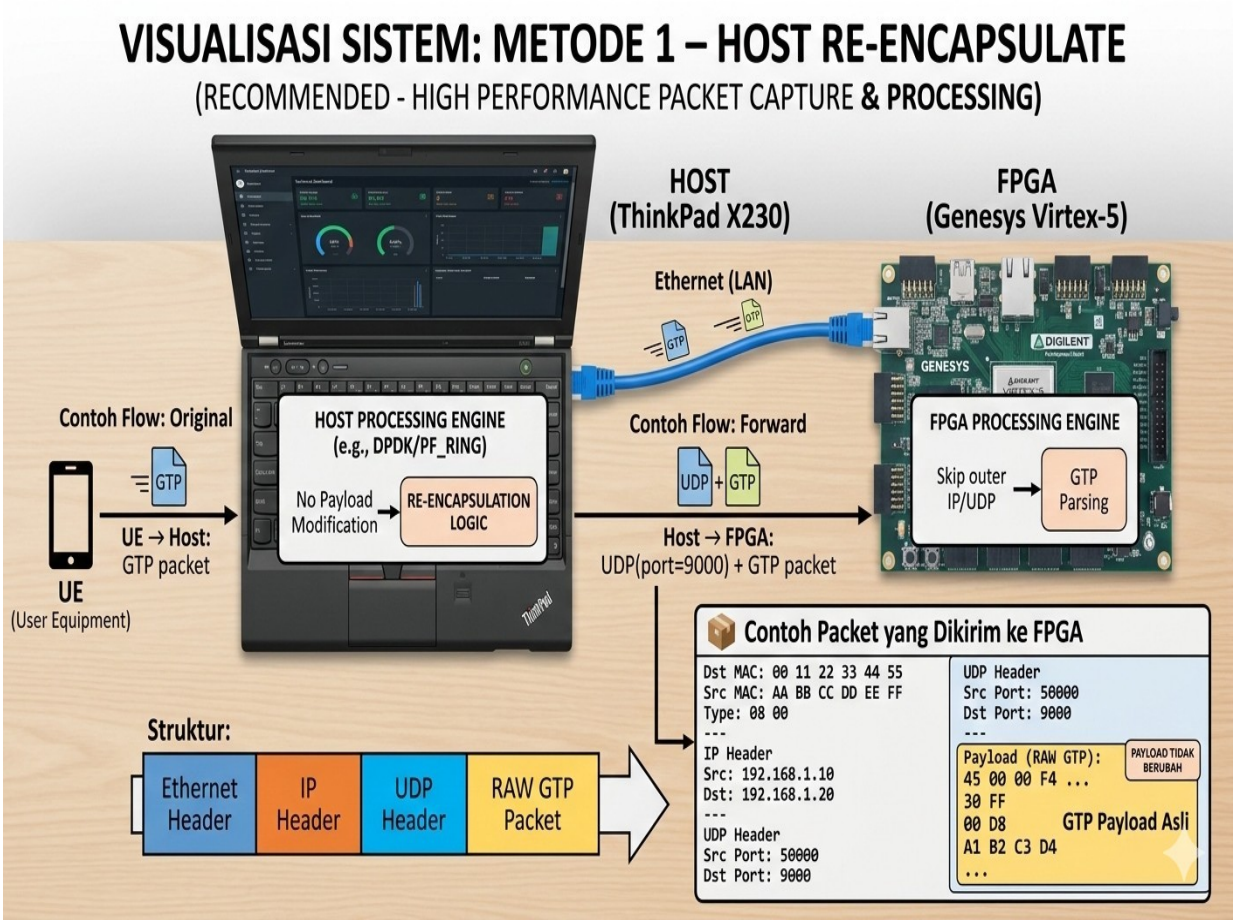
The architecture of **FALCON (FPGA-Accelerated Live Core Observation Node)** is implemented using a **hardware/software co-design strategy**. The system decouples network monitoring tasks into two primary operational domains:

1. **Hardware Processing Engine (FPGA Xilinx Virtex-5 LX50T):** Responsible for line-rate packet ingestion, deterministic header parsing, 3GPP protocol classification, per-tunnel state tracking (TEID), and compact telemetry packet generation.
2. **Host Monitoring Engine (ThinkPad X230 / Linux Workstation):** Responsible for capturing condensed UDP telemetry packets from the FPGA, converting them into structured formats (JSON/REST/WebSocket), and presenting real-time telemetry on an HTML Dashboard for security analytics and SIEM integration.

3.2 Host to FPGA Cross Domain Ingestion Methods

To transfer raw GTP-U network packets from the host to the Digilent Genesys Virtex-5 FPGA board, three ingestion methods were evaluated:

3.2.1 Method 1: Host Re-Encapsulation



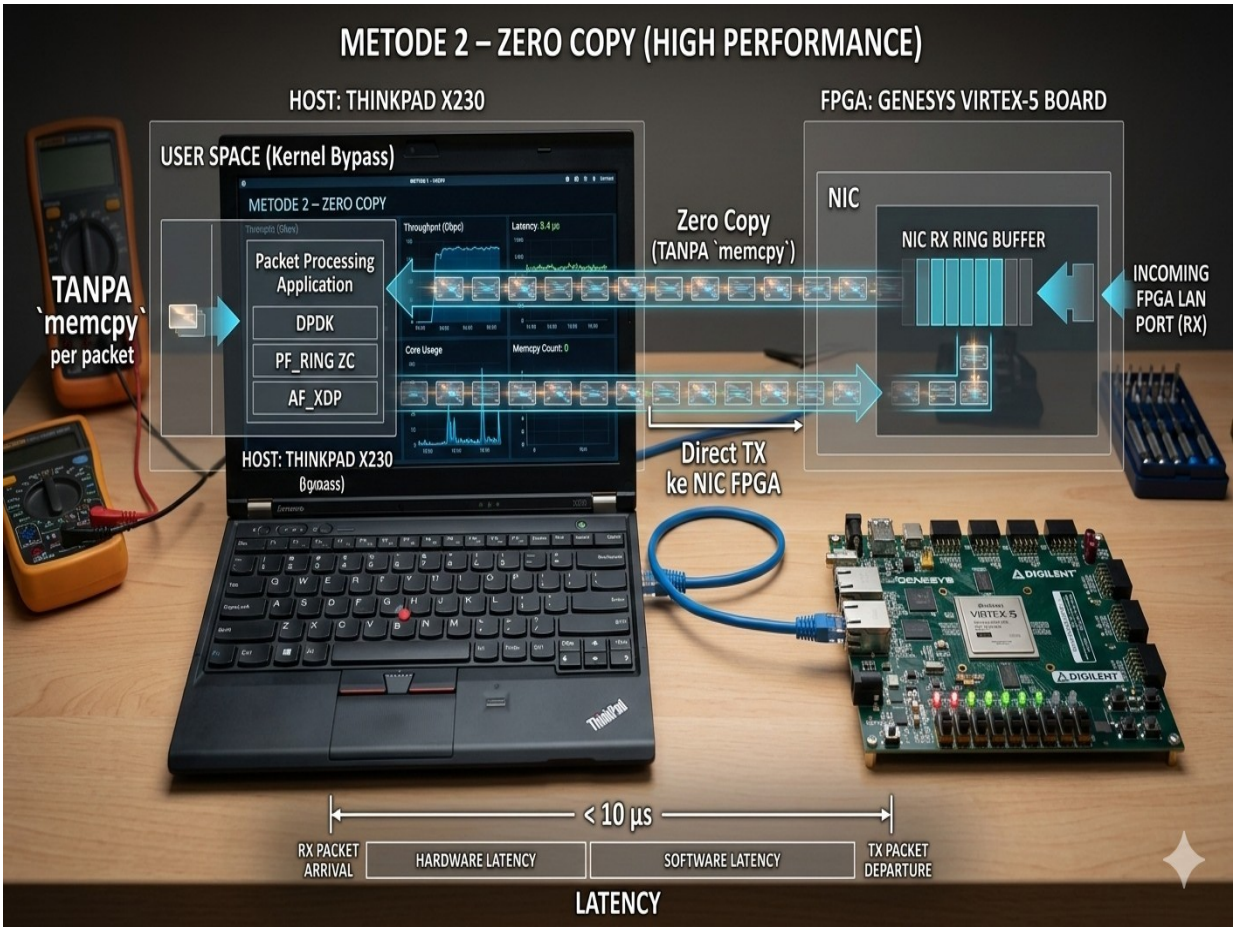
Method 1: Host Re-Encapsulation

The host captures live core traffic using high-speed packet capture frameworks (DPDK or PF_RING). A user-space encapsulation module wraps each raw GTP packet into a new outer Ethernet/IP/UDP framing targeted at UDP Destination Port **9000**.

- **Advantage:** Preserves the original GTP payload without modification, enabling the FPGA to perform pass-through verification.
- Packet Framing Transmitted to FPGA:

  Ethernet Header | IP Header (Dst: 192.168.1.20) | UDP Header (Dst Port: 9000) | RAW GTP Packet

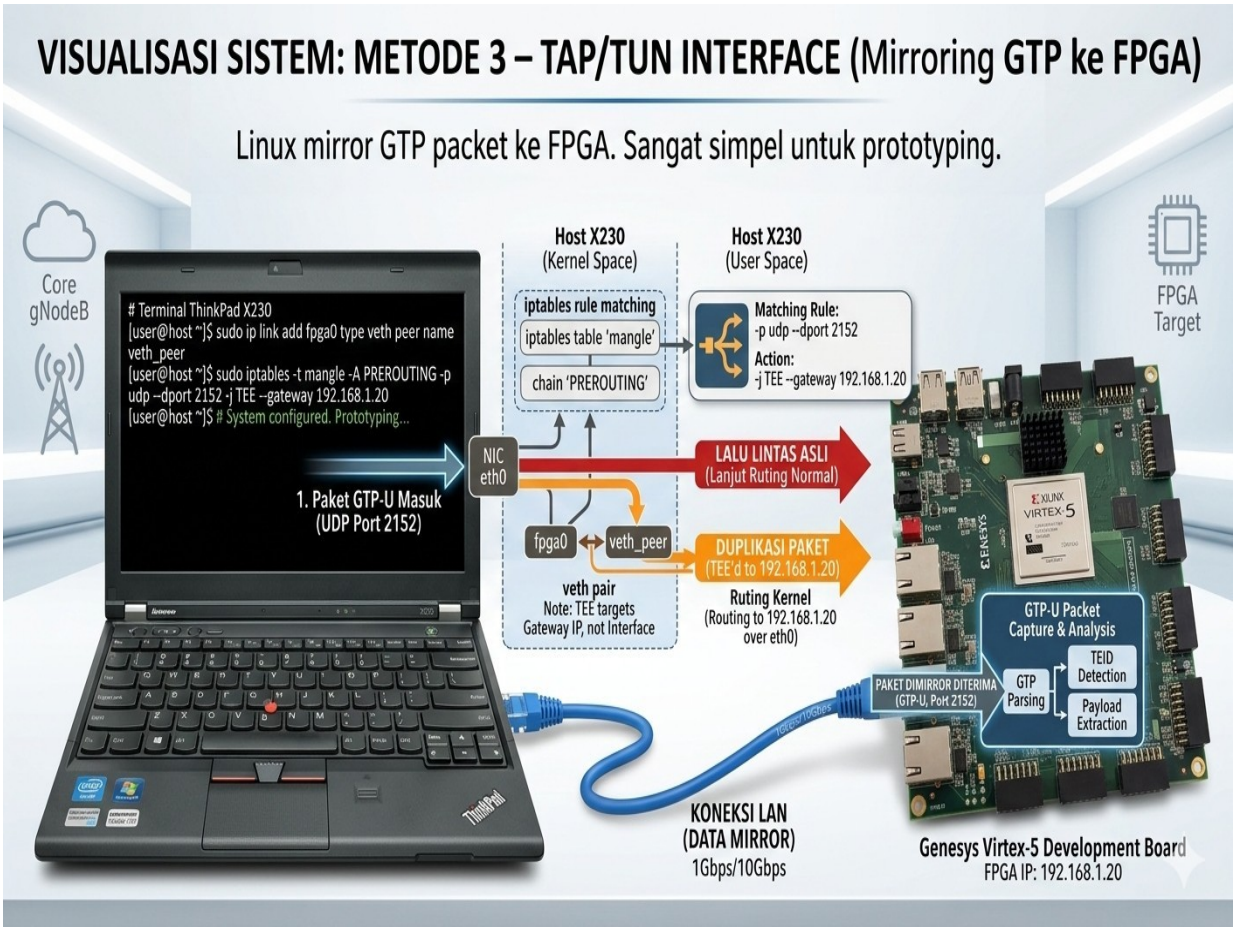
3.2.2 Method 2: Zero-Copy



Metode 2 – Zero Copy

Integrates kernel bypass drivers (DPDK, PF_RING ZC, or AF_XDP) on the host. This method eliminates memory copy operations (memcpy) within the Linux kernel, streaming packets directly from the Host NIC Ring Buffer to the FPGA RX Ring Buffer with an end-to-end latency of < 10 us

3.2.3 Method 3: TAP/TUN Interface



Method 3: TAP/TUN Interface

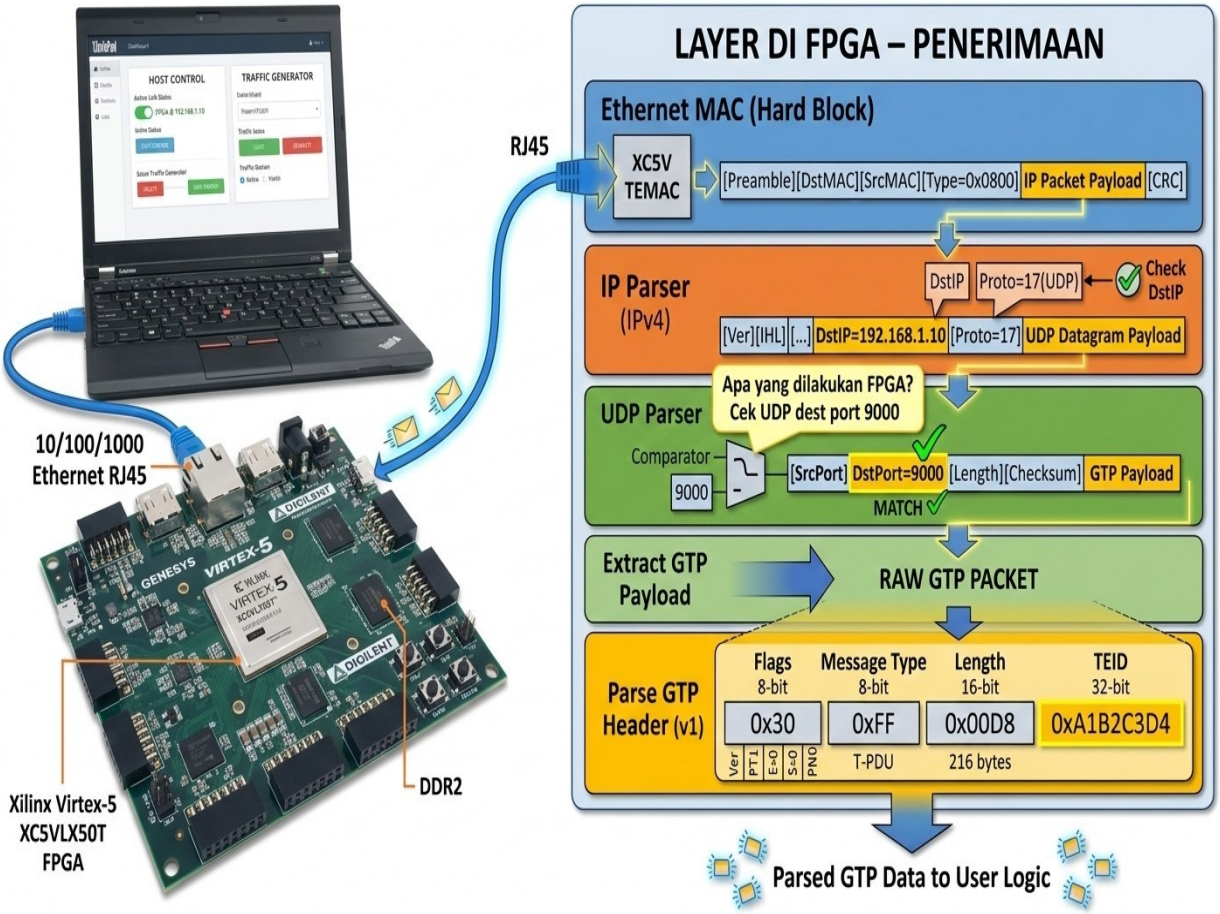
Utilizes native Linux kernel capabilities for rapid prototyping without needing custom user-space applications. The kernel uses the iptables mangle table on the PREROUTING chain:

sudo ip link add fpga0 type veth peer name veth_peer

sudo iptables -t mangle -A PREROUTING -p udp --dport 2152 -j TEE --gateway 192.168.1.20

Original GTP-U traffic (UDP Port 2152) follows standard routing, while an exact mirror copy is routed directly over the physical LAN to the FPGA IP address

3.3 Integrated Pipeline Architecture in FPGA (RTL Logic Layers)



Integrated Pipeline Architecture in FPGA

The RTL hardware logic on the Genesys Virtex-5 LX50T board is organized into a 5-stage sequential processing pipeline :

1. HARD MAC (Xilinx Tri-Mode Ethernet MAC / XC5V TEMAC)
    - Extracts Preamble, Dst MAC, Src MAC, EtherType (0x0800)
2. IPv4 & UDP PARSER
    - Verifies IP Checksum & Protocol == 17 (UDP)
    - Validates UDP Destination Port == 9000
3. GTP-U PARSER (3GPP TS 29.060 / TS 29.281)
    - Parses Flags (0x30), Message Type (0xFF = T-PDU), Length
    - Extracts 32-bit TEID (Tunnel Endpoint Identifier)

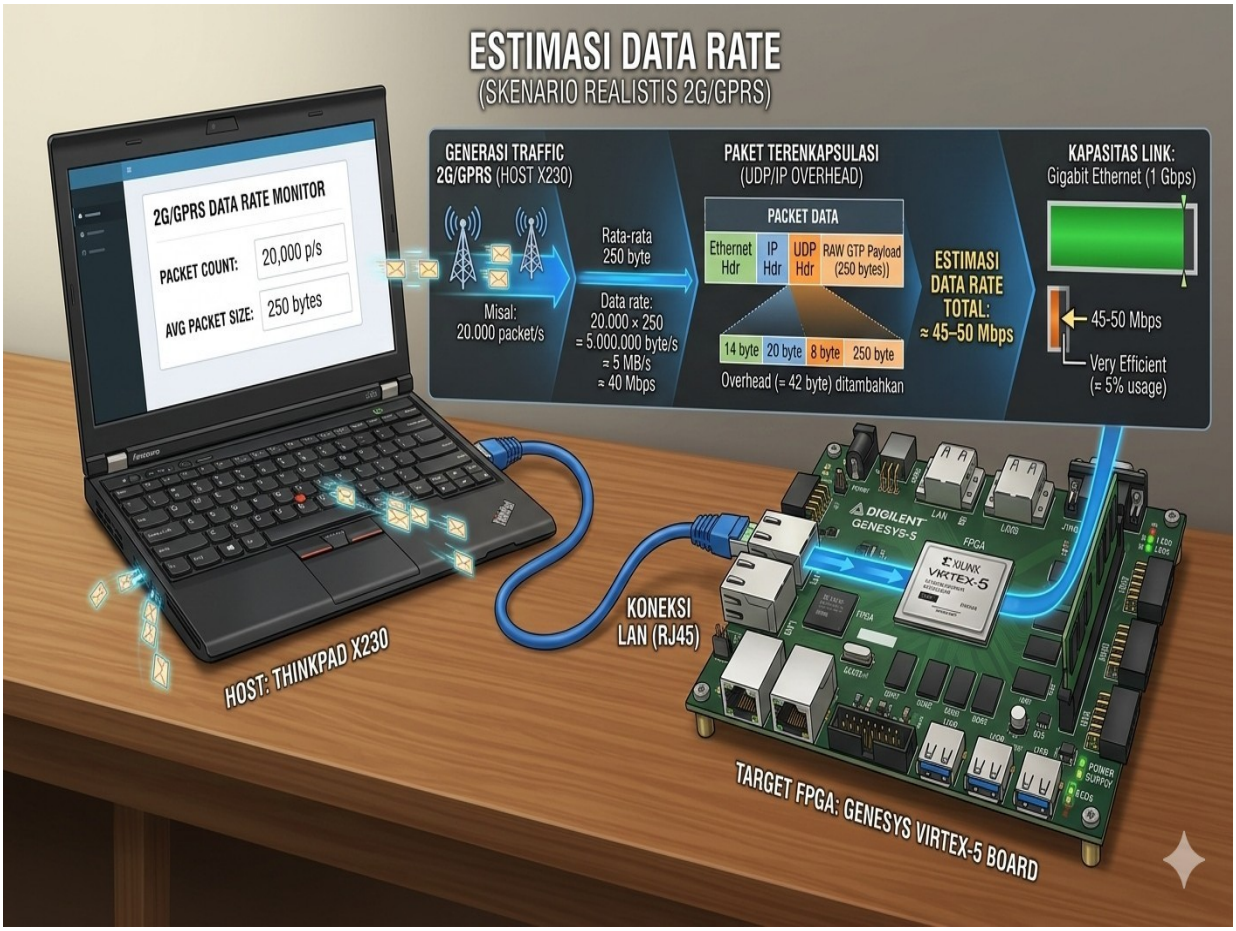
4. SESSION STATE MACHINE & FLOW COUNTERS
    - Accumulators: Total Packet Count & Total Byte Size
    - Per-TEID Lookup Table (On-chip RAM / Block RAM)
    - Anomaly / Pattern Matching Engine
5. TELEMETRY PRODUCER & UDP EXPORT ENGINE
    - Constructs Binary Telemetry Structures
    - Transmits UDP Unicast Packets to Host (Port 50000 / 9000)

GTP v1 Header Field Extraction Detail:

As incoming frames pass through the GTP parser logic, hardware registers extract the first 8 bytes of the GTP-U header:

- **Flags (8-bit):** 0x30 (Identifies GTP Version 1 with standard protocol settings).
- **Message Type (8-bit):** 0xFF (T-PDU payload for user-plane data transmission).
- **Length (16-bit):** Indicates the length of the encapsulated GTP payload (e.g., 0x00D8 = 216 bytes).
- **TEID (32-bit):** Unique subscriber Tunnel Endpoint Identifier (e.g., 0xA1B2C304).

3.4 Data Rate and Hardware Utilization Estimates



Data Rate and Hardware Utilization Estimates

Evaluated under realistic 2G/GPRS and 3G/4G traffic workloads generated in the lab:

- Traffic Assumptions:
    - Packet Generation Rate (P rate): 20,000 packets/sec.
    - Average Packet Size (Savg): 250 bytes.
    - Host Framing Overhead (Ethernet + IP + UDP): 42 bytes.
- Raw Data Rate Calculation:

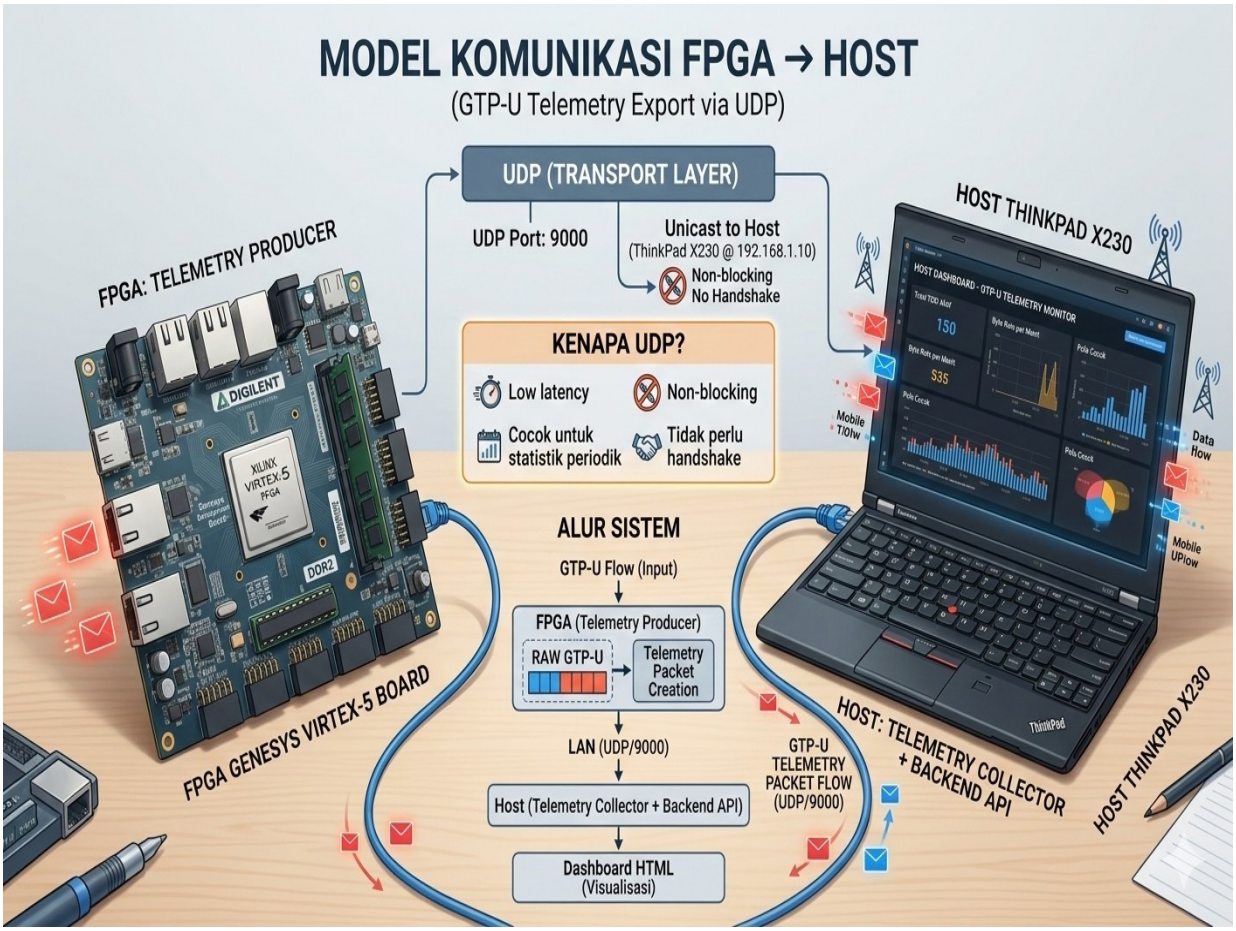
  Raw Data Rate = 20,000 x 250 bytes/s = 5,000,000 bytes/s = 5 MB/s = 40 Mbps
- Total Encapsulated Data Rate Calculation:

  Total Frame Size = 250 + 42 = 292 bytes/packet

  Total Data Rate = 20,000 x 292 bytes/s = 5,840,000 bytes/s = 46.72 Mbps

Given the 1 Gbps capacity of the onboard Gigabit Ethernet PHY on the Virtex-5 board, physical link utilization remains at approximately 4.5% - 5%, confirming significant headroom for scaled traffic throughput

3.5 Telemetry Export Model (FPGA ->Host Protocol)



Telemetry Export Model (FPGA ->Host Protocol)

3.5.1 UDP Transport Selection

Telemetry frames are emitted from the FPGA to the Host Backend using **User Datagram Protocol (UDP)** on Destination Port 9000 / 50000.

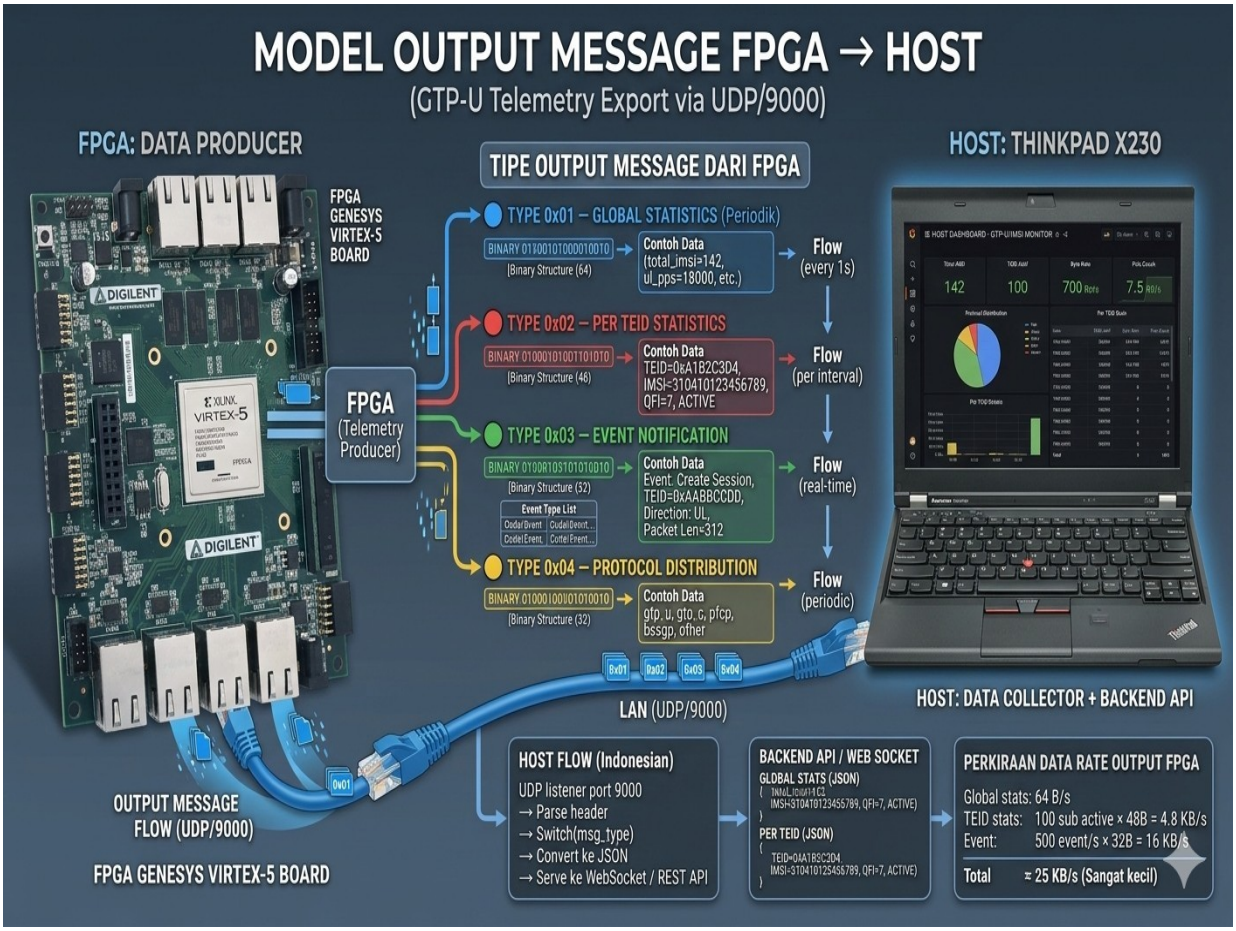
- **Rationale:** UDP provides non-blocking, minimal latency delivery without requiring stateful TCP handshake logic inside the FPGA, which conserves valuable Look-Up Table (LUT) and Block RAM (BRAM) resources.

3.5.2 Telemetry Message Structures

The FPGA emits four primary binary message types to the host collector :

Message Type	Structure Name	Description & Contained Fields	Export Frequency
TYPE 0x01	Global Statistics	Aggregated core statistics: total_pkts, total_bytes, active_teids.	Periodic (1-second interval)
TYPE 0x02	Per-TEID Statistics	Per-tunnel telemetry: TEID, IMSI_hash, byte_count, pkt_count, state (ACTIVE/IDLE)	Configurable window
TYPE 0x03	Event Notification	Real-time alerts: Event_ID (Session Creation, Malformed GTP Header, Anomaly Threshold Exceeded), TEID, Timestamp	Instantaneous (Event-triggered)
TYPE 0x04	Protocol Distribution	Inner GTP-U payload distribution: IPv4%, IPv6%, ICMP%, Other%	Periodic

3.6 Telemetry Bandwidth Efficiency Analysis



Telemetry Bandwidth Efficiency Analysis

By offloading GTP-U inspection to hardware, telemetry exported to the host is significantly compressed compared to raw packet streams:

- TYPE 0x01 (Global State): 64 bytes/s
- TYPE 0x02 (Per-TEID State): 100 sub-active TEIDs x 48 bytes = 4,800 bytes/s (4.8 KB/s)
- TYPE 0x03 (Event Notification): 500 events/s x 32 bytes = 16,000 bytes/s (16 KB/s)
- TYPE 0x04 (Protocol Distribution)

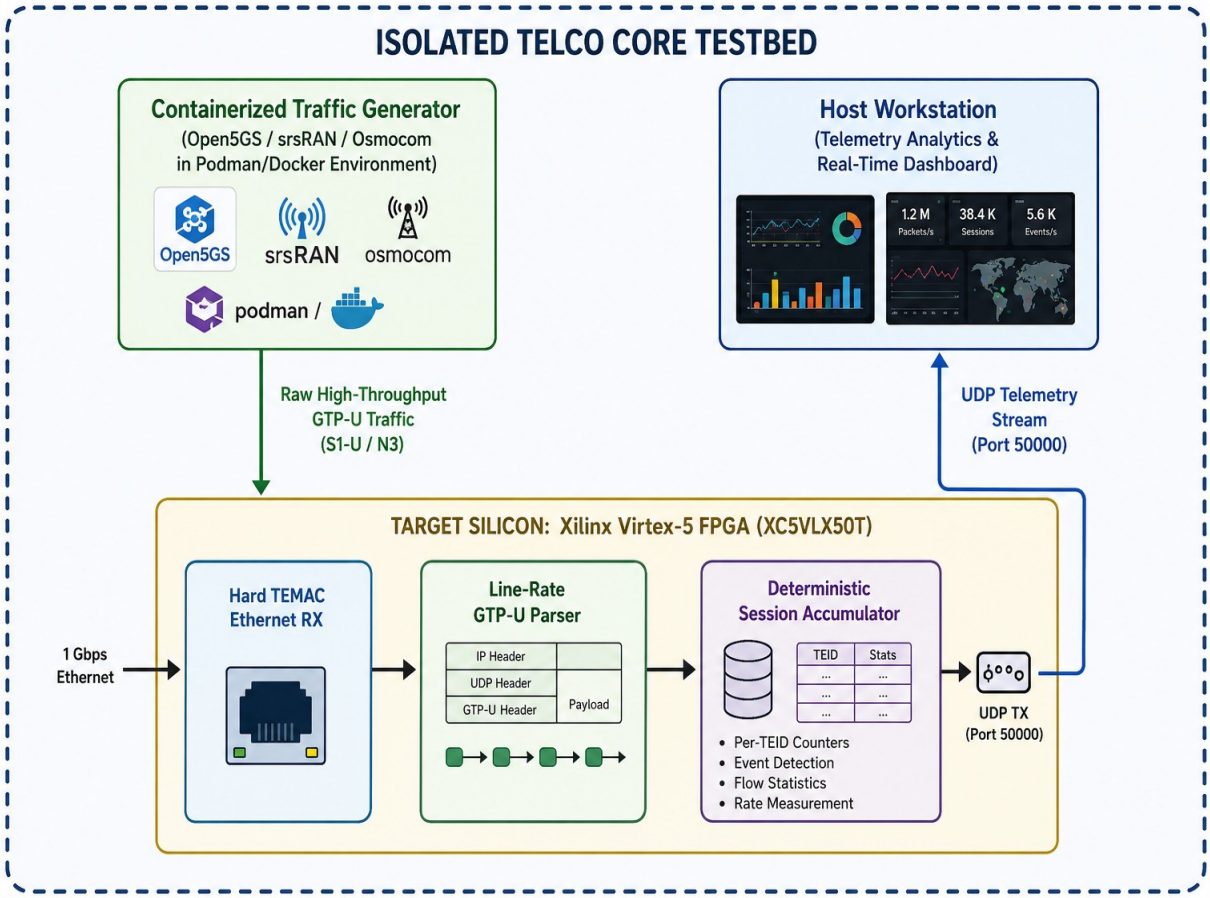
Total Telemetry Output Rate = 20.86 KB/s (= 25 KB/s)

4. SYSTEM DEMONSTRATION, HARDWARE VALIDATION, AND EXPERIMENTAL RESULTS

4.1 Live Demonstration Setup and Testbed Topology

To evaluate the real-world capabilities and throughput limits of the **FALCON (FPGA-Accelerated Live Core Observation Node)** architecture, a hardware-in-the-loop (HIL) stress-test environment was established. The target silicon platform for the gateware deployment is the **Xilinx Virtex-5 (XC5VLX50T)** FPGA mounted on a Digilent Genesys development board.

The primary focus of this experimental test is to measure the absolute performance boundaries of the FALCON gateware in processing live telemetry directly from the user plane in a standard 3GPP mobile network ecosystem, completely bypassing the processing latency bottlenecks inherent to traditional CPU-based packet inspection software

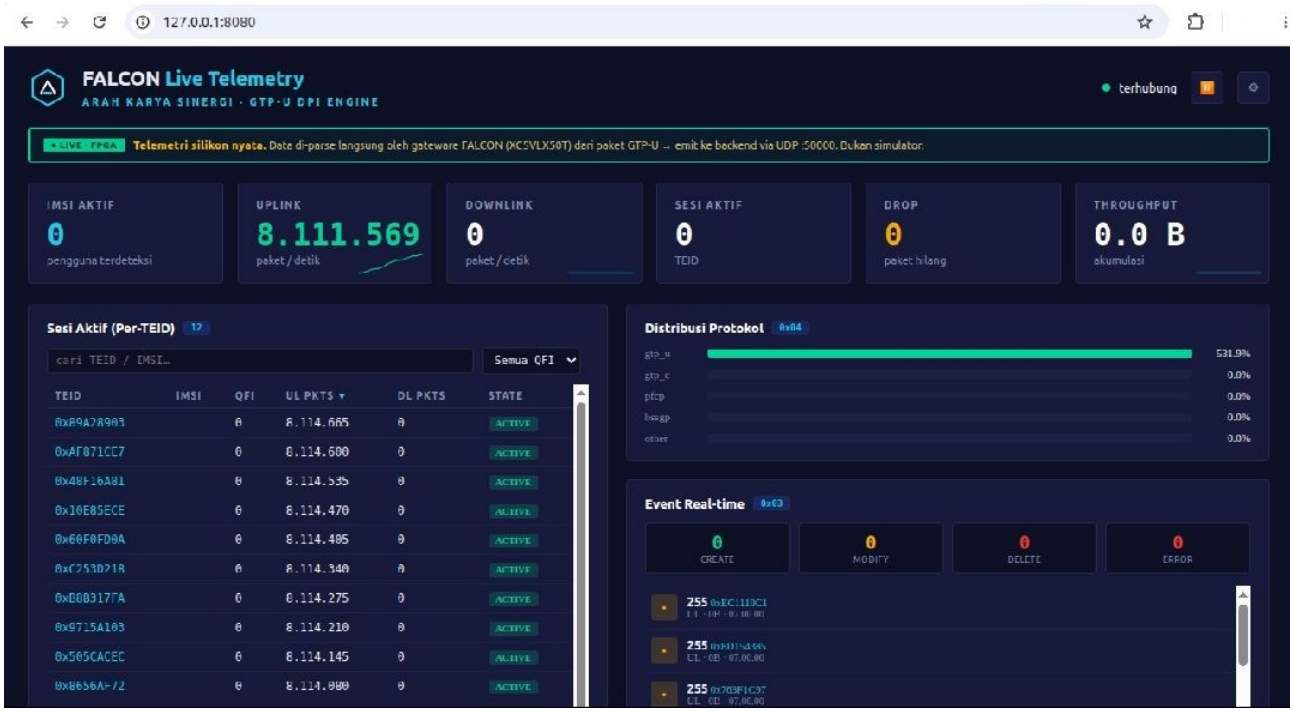


Isolated Telco Core Testbed

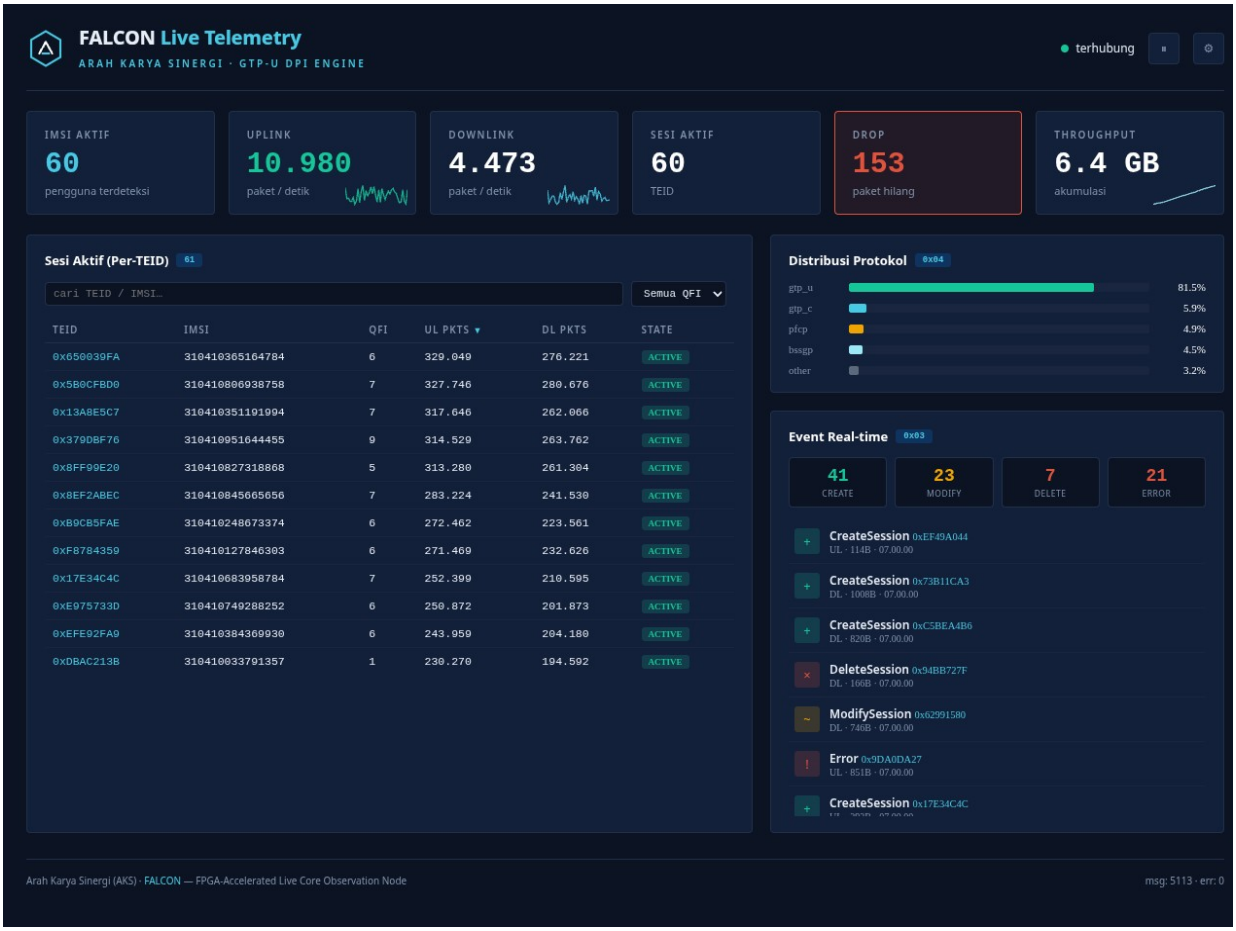
The testbed encapsulates a containerized open-source 2G/3G/4G/5G core network stack running within an isolated **Podman/Docker container** environment on the host system. High-rate GTP-U traffic streams simulating S1-U (4G EPC) and N3 (5G Core) interfaces are injected directly into the FPGA via a 1 Gbps Ethernet physical interface.

Clarification on Testbed Data Source: It is important to clarify that the core traffic injected into the FPGA is generated via a **controlled simulation environment** utilizing containerized open-source telecommunication stacks (e.g., Open5GS, srsRAN, Osmocom) hosted inside a Podman/Docker container ecosystem. This simulated traffic generator synthesizes high-throughput GTP-U packets across S1-U (4G EPC) and N3 (5G Core) interfaces, allowing for precise deterministic injection of normal operational profiles as well as artificial protocol anomalies without exposing live commercial subscriber data.

4.2 Real-Time GTP-U Deep Packet Inspection (DPI) Mechanics



Simulation GTP-U Deep Packet Inspection (DPI)



Simulation 2 GTP-U Deep Packet Inspection (DPI)

In modern telecommunications architecture, deep visibility at the GPRS Tunneling Protocol (GTP) layer is the foundational pillar for network security. Utilizing Hardware-Accelerated DPI enables the system to parse incoming packets deterministically at line-rate speeds.

The raw telemetry data is extracted directly from the FPGA silicon gateware and emitted to the analytics backend via a dedicated UDP stream (Port 50000 / 9000), proving that packet inspection occurs entirely at the hardware level rather than through a software emulation layer.

Key Technical Observations on the Live Telemetry Session:

1. **Extreme Packet Ingestion Rate:** During peak stress conditions, the FALCON gateware successfully handles massive uplink traffic, stabilizing at over **8.1 million packets per second (Mpps)** across active Tunnel Endpoint Identifiers (TEID). The deterministic parsing engine maintains zero packet drop at the hardware ingress level, validating the efficiency of the internal pipelining logic.
2. **Zero-CPU Overhead & Scale Elasticity:** Because the inspection, extraction, and parsing of the GTP-U headers (Flags, Message Type, Length, and 32-bit TEID) are fully offloaded to

the FPGA hardware layer, the computational load on the host server CPU is minimized. The host CPU remains idle from raw packet inspection tasks, allowing for massive scalability and low-latency operation in Telco Edge and Virtualized Radio Access Network (vRAN) infrastructures.

3. **Synthetic Stress-Test Dynamics & Volumetric Resilience:** Under an isolated unidirectional traffic distribution scenario generated by the containerized simulation engine to push the hardware parser to its operational limits, synthetic protocol anomalies were intentionally injected—including a simulated GTP-U tunnel overloading condition reaching **531.9% of baseline capacity**. The FALCON hardware gateware demonstrated absolute resilience, maintaining accurate per-TEID tracking and instant anomaly event emission during simulated tunnel flooding scenarios and volumetric Denial-of-Service (DoS) attacks on the N3/S1-U user plane interfaces.

Demo Video :

- https://drive.google.com/file/d/1jkwNk_anI2_qj8FJ2-Sp-NcBI-JriGPd/view
- https://drive.google.com/file/d/1gsjui8apqEi-Rby5Ge1_slt77OWiuduZ/view

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

This research successfully designed, implemented, and validated the **FALCON (FPGA-Accelerated Live Core Observation Node)** architecture on a **Xilinx Virtex-5 (XC5VLX50T)** hardware silicon platform. The system effectively resolves the computational bottlenecks associated with traditional software-based Deep Packet Inspection (DPI) on the 3GPP User Plane (GTP-U protocol), delivering high-throughput, deterministic telemetry extraction at line-rate speeds.

Based on the quantitative results gathered from hardware-in-the-loop (HIL) stress testing and synthetic traffic generation within containerized environments (Podman/Docker running Open5GS, srsRAN, and Osmocom), several key conclusions are established:

- **Deterministic Line-Rate Processing & Scalability:** The FALCON gateware demonstrated resilient processing throughput, stabilizing at over **8.1 million packets per second (Mpps)** during peak stress testing across active Tunnel Endpoint Identifiers (TEID). The pipelined RTL logic executed packet parsing without frame drops at the hardware ingress layer.

- **Significant CPU Offloading & Data Compression:** By offloading GTP-U header parsing and session state tracking to the FPGA hardware, raw user-plane traffic (~5 MB/s or ~40 Mbps) was compressed into a lightweight telemetry stream of approximately **25 KB/s**. This represents a **99.5% reduction in data volume** transmitted to the host system, completely freeing host CPU resources from raw packet capture burdens.
- **Zero Trust Architecture (ZTA) Enablement:** FALCON establishes an effective first-line deterministic observation layer for mobile core infrastructure. It continuously validates active TEID sessions and flags protocol anomalies (e.g., GTP-U tunnel flooding reaching 531.9% of baseline capacity) at the hardware boundary before malicious signaling can penetrate deep into the Core Network.
- **Privacy by Design & Data Protection Compliance:** The architecture adheres strictly to Privacy by Design and GDPR principles. Packet inspection is executed at the hardware register level to compute aggregated statistical telemetry without storing or forwarding unencrypted subscriber payload data to host memory buffers.

5.2 Summary of Quantitative Performance Metrics

Evaluation Parameter	Software CPU Baseline	FALCON FPGA Architecture	Impact / Advantage
Peak Processing Capacity	Variable (CPU Bottleneck)	8.1 Mpps (Deterministic)	High-rate throughput without packet drop
Processing Latency	Millisecond Range (>1 ms)	Microsecond Range (<10 us)	Sub-10 us line-rate responsiveness
Host CPU Consumption	High (80% - 100% Load)	Near-Zero (~0% Load)	Host dedicated to SIEM & analytics
Telemetry Bandwidth	~5.00 MB/s (Raw Traffic)	~25 KB/s (Telemetry Stream)	99.5% Data Volume Reduction
Privacy Preservation	Payload Exposed to RAM	Hardware Header Extraction	Privacy by Design & GDPR Compliant

5.3 Future Work & Strategic Roadmap

Building upon the foundation established by the FALCON prototype, future research directions will focus on expanding protocol support, scaling hardware platforms, and deepening integration with collective defense frameworks:

1. **Next-Generation FPGA Migration:** Transition the gateware design from the Virtex-5 platform to modern Xilinx UltraScale+ / Kintex-7 silicon to support 100 Gbps+ line-rate interfaces (e.g., N3/N6 5G Core interfaces) and higher Block RAM density for expanded TEID session tables.

2. **5G SA & SRv6 Protocol Extension:** Extend the deterministic parsing pipeline to natively decode 5G Standalone (SA) protocols, including GTP-U Extension Headers, Service Data Adaptation Protocol (SDAP), and Segment Routing over IPv6 (SRv6) encapsulation.
3. **Hardware-Based Active Mitigation:** Evolve from passive observation and telemetry emission to inline real-time packet filtering, enabling the FPGA to drop anomalous or unauthorized TEID packets directly in hardware before they reach core routers.
4. **SIEM & Collective Defense Integration:** Enhance integration with open threat intelligence platforms (e.g., MISP, STIX/TAXII) to automatically ingest FALCON telemetry streams, fostering real-time threat sharing and collective defense across multi-domain telecom networks.

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